Reactions of Organic Compounds

Answer key with worked solutions

A. Understanding

1. Reaction types.

> (a) **Addition:** two molecules form one, e.g. ethene + Br_2 → 1,2-dibromoethane. (b) **Substitution:** an atom or group is replaced, e.g. $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$. (c) **Elimination:** a small molecule is removed and C=C forms, e.g. ethanol → ethene + H_2O. (d) **Oxidation:** gain of O or loss of H, e.g. ethanol → ethanal. (e) **Condensation:** two molecules join and release water, e.g. acid + alcohol → ester + water.

B. Reaction types and products

2. Classification.

> (a) **addition** (b) **substitution** (c) **elimination** (d) **oxidation** (e) **condensation** (f) **substitution**

3. Products.

> (a) **butane** (b) **2,3-dibromobutane**, $CH_3CHBrCHBrCH_3$ (c) **ethanol** (d) **propanoic acid** (reflux takes the oxidation all the way) (e) **butanone** (secondary alcohol → ketone)

C. Tests and selectivity

4. Tests.

> (a) Shake with **bromine water**: hex-1-ene turns it from **orange-brown to colourless**; hexane leaves it orange. (b) Warm with **acidified potassium dichromate**: propan-1-ol (primary) turns it from **orange to green**; 2-methylpropan-2-ol (tertiary) leaves it **orange**.

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5. Aldehyde or acid.

> To get **propanal**, use a limited amount of oxidizing agent and **distil** the aldehyde off as it forms (it boils lower than the alcohol), so it is not oxidized further. To get **propanoic acid**, use excess oxidizing agent and **heat under reflux**, so the aldehyde stays in the flask and is oxidized to the acid.

6. Propene + HBr.

> (a) **1-bromopropane** and **2-bromopropane**. (b) **2-bromopropane** is the major product (Markovnikov's rule): the H adds to the end carbon, which already has more hydrogens, and Br adds to the middle carbon.

D. Calculations

7. Ethanol from ethene.

> $n(\mathrm{C_2H_4}) = \dfrac{1.40 \times 10^4\ \mathrm{g}}{28.05\ \mathrm{g/mol}} = 499.1\ \mathrm{mol}$; 1 : 1 ratio, so $n(\text{ethanol}) = 499.1\ \mathrm{mol}$
> Theoretical yield $= 499.1\ \mathrm{mol} \times 46.07\ \mathrm{g/mol} = 2.299 \times 10^4\ \mathrm{g} =$ **23.0 kg**
> Percent yield $= \dfrac{21.2\ \mathrm{kg}}{22.99\ \mathrm{kg}} \times 100\ \% =$ **92.2 %**

8. Ethanoic acid.

> $n(\text{ethanol}) = \dfrac{9.21\ \mathrm{g}}{46.07\ \mathrm{g/mol}} = 0.1999\ \mathrm{mol}$; 1 : 1 ratio (
> $\mathrm{CH_3CH_2OH + 2\,[O] \longrightarrow CH_3COOH + H_2O}$)
> $m = 0.1999\ \mathrm{mol} \times 60.05\ \mathrm{g/mol} =$ **12.0 g**

E. Challenge

9. Synthesis of ethyl ethanoate.

> 1. Ethene + steam, acid catalyst → **ethanol** (**addition**, hydration). 2. Oxidize part of the ethanol with **excess acidified potassium dichromate, heated under reflux → ethanoic acid** (**oxidation**). 3. Warm ethanoic acid with the rest of the ethanol and a few drops of **concentrated sulfuric acid → ethyl ethanoate** + water (**condensation**, esterification).

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